

PROFILE

Third-year Computer Science student at McGill University with experience in full-stack development, DevOps workflows, and building reliable software for real-world environments. Developed TypeScript/JavaScript applications with backend integration, fault-tolerant session handling, and automated testing, supporting healthcare and manufacturing use cases. Background in statistics and portfolio optimization, with a focus on translating business and analytical requirements into maintainable, data-driven solutions.

WORK EXPERIENCE

3AGE (SABANCI UNIVERSITY INNOVENT) — AI & SOFTWARE DEVELOPMENT INTERN *December 2025 - Ongoing*

- Developed a state-driven kiosk platform supporting **structured healthcare decision workflows**
- Implemented **fault-tolerant session logic** (timeouts, resets, inactivity detection) for real-world kiosk environments
- Planned end-to-end **UI flows** and built **React/TypeScript** interfaces for clinical workflows

BSH HAUSGERÄTE — ROBOTICS AUTOMATION INTERN, MESA TEAM *July - August 2024*

- Optimized manufacturing workflows using **KUKA & FANUC** systems, improving process stability and cycle efficiency
- Built **100+ automated test suites (JUnit)** to validate control logic and prevent regressions in live production
- Diagnosed bottlenecks and deployed updates in a time-critical operational environment, strengthening reliability

THE GAME CIRCLE — GAME DEVELOPMENT INTERN *June – July 2022*

- Analyzed and refined **sound effect** pipelines for mobile game prototypes in collaboration with Prof. Dr. Veysi İşler
- Designed and adjusted **3D visual elements** using **Blender** to enhance in-game interactions
- Optimized gameplay mechanics by **analyzing player feedback** and ensuring audio-visual alignment

PROJECTS - [nehirozsunar.com/projects](#)

ANTONIO — AI VOICE CAR ASSISTANT *Live Link: [myantonio.app](#)*

- Built a **voice-activated AI car assistant** with personality modes and real-time Spotify & Google Maps integration
- Implemented **wake word** detection (Porcupine SDK) and a **TTS** priority queue with a hybrid **iOS/desktop** audio architecture

HEALTH HUB KIOSK INTERFACE *GitHub: [nehirozs/health_hub-frontend](#)*

- Developed full-stack healthcare application using **React** with **backend API** integration for health screening workflows
- Implemented robust session management with timeout handling, automatic resets, and inactivity detection for kiosk environments

QUANTITATIVE PORTFOLIO OPTIMIZATION FRAMEWORK *GitHub: [nehirozs/MPT-Risk-Analysis](#)*

- Implemented **Modern Portfolio Theory** with **Sharpe ratio optimization**, efficient **frontier analysis**, and backtesting strategies
- Built **Monte Carlo** simulation engine and **Black-Litterman** model integration for risk assessment and market view incorporation
- Technologies: **Python (NumPy, Pandas, Matplotlib)**, statistical modeling, time-series analysis

EDUCATION

MCGILL UNIVERSITY *Montreal, Canada* *B.Sc. in Computer Science (Minor in Economics) August 2023 - April 2027 (expected)*

- Advanced standing granted through AP credits; currently in third year of studies.
- Relevant Coursework: Operating Systems (**C**), Algorithm Design, Algorithms and Data Structures (**Java**), Database Systems (**SQL, ER Design**), Software Design (**Java, OOP**), Programming Languages & Paradigms (**OCaml**), Computer Systems (**C, MIPS, Git**), Programming Challenges (Competitive Programming), Probability, Linear Algebra, Discrete Structures, Statistics

ROBERT COLLEGE OF ISTANBUL *Istanbul, Turkey* *High School Diploma 2018 - 2023*

MIMAR SINAN FINE ARTS UNIVERSITY *Istanbul, Turkey* *2020 – 2022*

- Ballet Conservatory (top 3 of 64 students, graduated 2 years early)*
- Currently training with Les Grands Ballets Canadiens

SKILLS & LANGUAGES

- Programming:** Python (NumPy, Pandas), Node.js, Java, C, JavaScript, TypeScript, HTML/CSS, React, OCaml, Bash, SQL
- Systems & Tools:** Linux/Unix, Git, GitHub/GitLab, JUnit, MIPS Assembly, Logisim, Blender
- Languages:** English (Advanced), Turkish (Native), French (Intermediate)